*ms*2: A molecular simulation tool for thermodynamic properties, release 3.0

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Abstract

A new version release (3.0) of the molecular simulation tool *ms*2 [Deublein et al., Comput. Phys. Commun. 182 (2011) 2350 and Glass et al., Comput. Phys. Commun. 185 (2014) 3302] is presented. Version 3.0 of *ms*2 features two additional ensembles, i.e. microcanonical (NVE) and isobaric–isoenthalpic (NpH), various Helmholtz energy derivatives in the NVE ensemble, thermodynamic integration as a method for calculating the chemical potential, the Maxwell-Stefan diffusion coefficients of quaternary mixtures, statistics for sampling hydrogen bonds, the osmotic pressure for calculating the activity of solvents, the smooth-particle mesh Ewald summation, as well as the ability to carry out molecular dynamics runs for an arbitrary number of state points in a single program execution.

Keywords: molecular simulation, molecular dynamics, Monte Carlo

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New version program summary

Program Title: *ms*2

Operating system: Unix/Linux

Licensing provisions: Special license supplied by the authors

Programming language: Fortran95

Has the code been vectorized or parallelized: yes (Message Passing Interface (MPI) protocol and OpenMP scalability is up to 2000 cores)

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Distribution format: tar.gz

Classification: 7.7, 7.9, 12

Catalogue identifiers of previous versions: AEJF_v1_0 and AEJF_v2_0

Supplementary material:

Journal reference of previous versions: Deublein et al., Comput. Phys. Commun. 182 (2011) 2350 and Glass et al., Comput. Phys. Commun. 185 (2014) 3302

Does the new version supersede the previous version?: Yes

Reasons for the new version: Introduction of new features as well as enhancement of computational efficiency

Summary of revisions: Two new ensembles (NVE and NpH), new properties (Helmholtz energy derivatives, chemical potential via thermodynamic integration, Maxwell-Stefan diffusion coefficients of quaternary mixtures, activity coefficients via osmotic pressure), new functionalities (detection of hydrogen bonding, the smooth-particle mesh Ewald summation, the ability to carry out molecular dynamics runs for an arbitrary number of state points in a single program execution)

Nature of problem: Calculation of application oriented thermodynamic properties: vapor-liquid equilibria of pure fluids and multi-component mixtures, thermal, caloric and entropic data as well as transport properties

Solution method: Molecular dynamics, Monte Carlo, various ensembles, grand equilibrium method, Green-Kubo formalism, Lustig formalism, OPAS method, smooth-particle mesh Ewald summation

Typical running time: Varies (typically from a couple of hours up to days), depending on the specific scenario (system size, calculated properties, number of CPU cores used)

Restrictions: Typical problems addressed by $ms2$ are solved by simulating systems containing typically $1000 - 5000$ molecules that are modelled as rigid bodies

Documentation: Documentation is provided with the installation package and is available at http://www.ms-2.de

1. Introduction

The software $ms2$ is a molecular force field based simulation tool that allows for the calculation of thermodynamic properties of pure substances and mixtures in their homogeneous states as well as vapor-liquid equilibria. The tool was first released in 2011 [?] that was followed by a major update ($ms2$ release 2.0) in 2014 [?]. The following chapters introduce and describe the new features of the present update, i.e. release 3.0.

2. Microcanonical and isobaric-isoenthalpic ensembles

An ensemble is the set of all theoretically possible microscopic configurations on the molecular level under specific macroscopic constrains. The microcanonical (NVE) ensemble is the set of all configurations that fulfill the condition of having the same number of particles N, volume V and energy E, whereas the isobaric-isoenthalpic (NpH) ensemble represents a system at constant number of particles N, pressure p and enthalpy H. Molecular dynamics (MD) mimics the time evolution of the system by numerically solving Newton's equations of motion for all considered molecules. Because of the nature of this solution, the time is discredized and the method yields microscopic configurations at discrete and consecutive time steps. The Monte Carlo (MC) method is the application of statistical mechanics to describe molecular systems. With this approach, microscopic configurations are generated by random numbers that are potentially accepted such that only relevant and physically meaningful configurations are sampled [?]. The generation and acceptance of these configurations is governed by specific probabilities that are ensemble-dependent and described in the literature for essentially every relevant ensemble. For MC simulations, the NVE and NpH ensembles were implemented in $ms2$ 3.0 according to Refs. [? ?]. For MD simulations, the pressure is kept constant using Andersen's barostat [?] in case of the NpH ensemble. Because the solution of Newton's equations of motion is approximate, the total energy of the system $E = K + U$, which consists of a kinetic K (exclusively molecular velocity dependent) and potential U (exclusively molecular position dependent) energy contribution, is not rigorously conserved in a standard MD run. Therefore, the translational and rotational velocity of every molecule is rescaled such that the total kinetic energy K fulfils $K = E - U$, where the total energy E is specified and the potential energy U is calculated from the current configuration. For a NpH ensemble run, the solution is analogous: Velocities are rescaled such that the current kinetic energy K fulfils $K = H - U - pV$, where H and p are specified, U and V are dependent on the current microscopic configuration. This extends the ensembles available in $ms2$ to five: NVT, NVE, NpT, NpH and μVT, where μ is the chemical potential.
For verification purposes, Table 1 contains numerical results for methyl fluoride modeled by a two-center Lennard–Jones potential [?] at $T = 300$ K and $\rho = 1$ mol/l. Figure 1 shows the running averages of the calculated properties in Table 1 at the same state point.

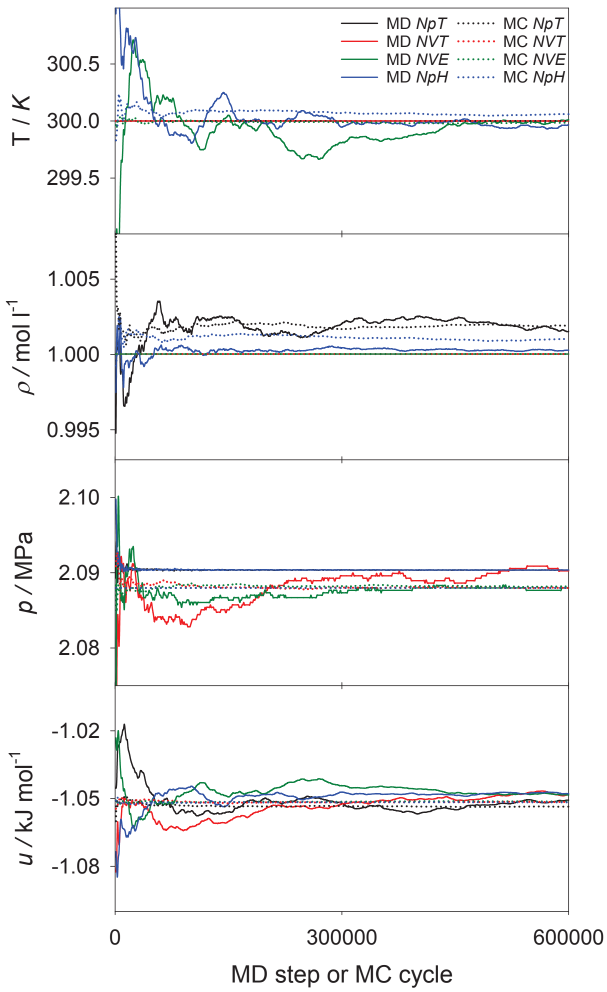

Figure 1: Comparison of the ensembles implemented in *ms*2 for methyl fluoride (R41)[**?**] in terms of running averages for temperature T, density ρ, pressure p and potential energy u.

Table 1: Comparison of the ensembles implemented in $ms2$ for methyl fluoride (R41) [**?**] in terms of temperature T, density ρ, pressure p and potential energy u. Numbers in brackets denote statistical uncertainties in the last digit that were estimated with the method of Flyvbjerg and Petersen [**?**].

		T K	ρ mol/l	p MPa	u kJ/mol
MD	NVT	300	1	2.090(2)	-1048(3)
	NpT	300	1.0015(7)	2.090	-1051(3)
	NVE	300.0(1)	1	2.088(1)	-1048(3)
	NpH	300.0(1)	1.0003(2)	2.090	-1048(2)
MC	NVT	300	1	2.0880(2)	-1051.6(2)
	NpT	300	1.0019(2)	2.0880	-1053.6(3)
	NVE	299.99(1)	1	2.0882(2)	-1051.3(2)
	NpH	300.06(2)	1.0010(1)	2.0880	-1051.7(2)

3. Helmholtz energy derivatives in the microcanonical ensemble

The generalized calculation of the Helmholtz energy derivatives

$$A^{\mathrm{r}}_{nm} = (1/T)^n \rho^m \frac{\partial^{n+m} f^{\mathrm{r}}(T,\rho)/(RT)}{\partial(1/T)^n \partial\rho^m}, \tag{1}$$

with the molar residual Helmholtz energy f^{r}, the temperature T, the density ρ and the molar gas constant R was introduced in $ms2$ also for the NVE ensemble up to the order $n = 3$ and $m = 2$. It can be shown that the total reduced Helmholtz energy $f/(RT)$ can be additively separated into an ideal $f^{\mathrm{o}}/(RT)$ and a residual contribution $f^{\mathrm{r}}/(RT)$. The ideal contribution by definition corresponds to the value of $f/(RT)$ when no intermolecular interactions are at work [**?**]. $f^{\mathrm{o}}/(RT)$ consists of an exclusively temperature and an exclusively density dependent part. The latter has a trivial density dependence $\ln(\rho/\rho_{\mathrm{ref}})$, whereas the former is non-trivial. A formalism that allows for the calculation of these derivatives on the fly from a single simulation run per state point was published by Lustig [**? ?**] and was already introduced for the NVT ensemble in the previous release. $ms2$ 3.0 now yields these derivatives also for NVE simulations. Here, however, in contrast to NVT simulations, there are two numerical results for each calculated derivative A_{nm} due to the two possible entropy definitions in statistical mechanics [**?**]. In any case, these two different sets of results must be identical in the thermodynamic limit ($N \longrightarrow \infty$). In practice, they already agree within their statistical uncertainty for a simulation with around a thousand molecules. For verification purposes, Table 2 contains numerical results for methyl fluoride [**?**].
A detailed description of how to convert these derivatives into common thermodynamic properties is provided in the supplementary material of release 2.0 of $ms2$ [**?**] and in Ref. [**?**].

4. Thermodynamic integration

The previously described method of Lustig does not allow for the direct sampling of the chemical potential, i.e. entropic properties like A_{00}. Such an effort requires techniques based

Table 2: Comparison of the residual Helmholtz energy derivatives A^{r}_{nm} of methyl fluoride (R41) [?] at $T = 300$ K and $\rho = 1$ mol/l. Numbers in brackets denote statistical uncertainties in the last digit that were estimated with the method of Flyvbjerg and Petersen [?]. The superscripts (1) and (2) indicate two different entropy definitions [?].

		A^{r}_{10}	A^{r}_{01}	A^{r}_{20}	A^{r}_{11}	A^{r}_{02}
	NVT	-0.420(1)	-0.1620(6)	-0.54(4)	-0.40(2)	0.03(3)
MD	NVE^1	-0.418(1)	-0.1622(4)	-0.55(3)	-0.37(1)	0.05(3)
	NVE^2	-0.418(1)	-0.1621(4)	-0.55(3)	-0.37(1)	0.05(3)
	NVT	-0.42158(8)	-0.16290(8)	-0.562(2)	-0.391(2)	0.011(6)
MC	NVE^1	-0.42149(5)	-0.16280(8)	-0.560(2)	-0.385(3)	0.024(6)
	NVE^2	-0.42133(5)	-0.16274(8)	-0.560(2)	-0.385(3)	0.024(6)

on free energy calculation, such as particle insertion and/or thermodynamic integration [?]. Widom's particle insertion method [?] is a conceptually straightforward approach to calculate the chemical potential with low computational cost, both for pure substances and mixtures. The total chemical potential μ_i of species i can be separated into an ideal (o) and a residual (r) contribution in the same way as the Helmholtz energy is decomposed (cf. section 2): $\mu_i(T, \rho, x_i) = \mu_i^{\mathrm{o}}(T) + RT \ln(N_i/(V\rho_{\mathrm{ref}})) + \mu_i^r(T, \rho, x_i)$, where N_i is the number of molecules of species i, $\rho = N/V$, $x_i = N_i/N$ and ρ_{ref} is an arbitrary reference density. The part $\mu_i - \mu_i^{\mathrm{o}}(T)$ is often referred to as the configurational chemical potential μ_i^{conf}. Widom's method requires the frequent insertion of an additional ($i = N + 1$) test particle into the simulation volume at a random position with a random orientation. At constant temperature and constant pressure or volume the potential energy U_i of this test particle, i.e. the interaction energy with all other "real" N molecules, yields the configurational chemical potential according to

$$\mu_i^{\mathrm{conf}} = \mu_i - \mu_i^{\mathrm{o}}(T) = -k_{\mathrm{B}} T \ln \frac{\langle V \exp(-U_i/k_{\mathrm{B}}T)\rangle}{\langle N_i \rangle}, \tag{2}$$

where k_{B} is Boltzmann's constant. The test particle is removed immediately after the calculation of its potential energy U_i, thus it does not influence the real molecules in the system in any way. In contrast to the usual convention, the brackets <> have a dual meaning here: They stand for either NVT or NpT ensemble averages as well as an integral over all possible positions and orientations of the test particles added to the system. The density of the system has a significant influence on the accuracy of this method. However, for state points with a high density, test particles almost always overlap with real molecules, which leads to a potential energy $U_i \to \infty$ and thus to a vanishing contribution to Eq. (2), resulting in poor statistics and often even to complete failure of sampling.

Thermodynamic integration is one solution to overcome the limitations of Widom's particle insertion method. The idea behind calculating the chemical potential by thermodynamic integration is to avoid insertion of the test particle in a challenging system A, but rather perform it in system B for which this can be done without sampling problems. Since the chemical potential is a state variable, its difference $\mu_{A,i} - \mu_{B,i}$ can be calculated along any path between states A and B, which

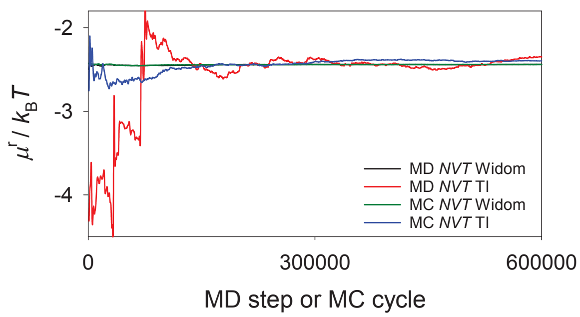

Figure 2: Comparison for the residual chemical potential μ^r of methyl fluoride (R41) [**?**] at $T = 300$ K and $\rho = 20$ mol/l calculated by $ms2$ using either Widom's particle insertion method (Widom) or the present thermodynamic integration method (TI).

is represented by the scalar parameter λ. It can be shown [**?**] that the relation between λ and $\mu_{A,i} - \mu_{B,i}$ is

$$\mu_{A,i} - \mu_{B,i} = \int_{B(\lambda_{\min})}^{A(\lambda_{\max})} \left\langle \frac{\partial U_i(\lambda)}{\partial \lambda} \right\rangle d\lambda \tag{3}$$

for particle i. The bracket $<>$ in this equation denotes NVT or NpT ensemble averages and U_i is the potential energy of particle i that must be a part of the system the same way as the other molecules are. The only difference between particle i and all other molecules is that its interaction energy $U_i(\lambda)$ is scaled between states $A(\lambda_{\max})$ and $B(\lambda_{\min})$ with λ. As long as $\partial U_i(\lambda)/\partial\lambda$ can be calculated analytically and sampled during simulation, the actual way of scaling $U_i(\lambda)$ with λ can be chosen arbitrarily because μ_i is a state variable. The integration with respect to λ is carried out numerically. Assuming that $\mu_{B,i}$ is practically zero or at least can be successfully calculated by Widom's particle insertion method for state B using Eq. (2), $\mu_{A,i}$ yields the configurational part of the total chemical potential. The non-linear scaling $U_i(\lambda) = \lambda^d U_i$ for $\lambda_{\min} \leq \lambda \leq 1 = \lambda_{\max}$ with an adaptive sampling technique [**?**] was implemented for MC simulations with d and $\lambda_{\min}$ being input parameters. This adaptive technique allows for the sampling of the entire range of $\lambda_{\min} \leq \lambda \leq 1$ in a single MC run with an arbitrary resolution for numerical integration. In addition to the standard NVT and NpT ensemble MC moves, the simulation includes changes of λ controlled by a proper MC acceptance criterion, ensuring visits at each discrete λ value in the range $\lambda_{\min} \leq \lambda \leq 1$. For MD simulations, changes of λ are also carried out on the fly in a single simulation, but without any acceptance criterion. A detailed description of the parameter setup is given in the supplementary material. Figure 2 shows the running averages for the residual chemical potential of methyl fluoride (R41) at $T = 300$ K and $\rho = 20$ mol/l calculated by either Widom's test particle insertion or the present thermodynamic integration method.

5. Quaternary Maxwell-Stefan diffusion coefficients

*ms*2 employs the Green-Kubo formalism based on the net velocity correlation function to obtain $n \times n$ phenomenological coefficients [?]

$$L_{ij} = \frac{1}{3N} \int_0^\infty \left\langle \sum_{k=1}^{N_i} \mathbf{v}_{i,k}(0) \cdot \sum_{l=1}^{N_j} \mathbf{v}_{j,l}(t) \right\rangle dt, \tag{4}$$

in a mixture of n components. Here, N is the total number of molecules, N_i is the number of molecules of species i and $\mathbf{v}_{i,k}(t)$ denotes the center of mass velocity vector of the k-th molecule of species i at time t. Note that the phenomenological coefficients given in Eq. (4) are constrained by [?]

$$\sum_i M_i L_{ij} = 0\,, \tag{5}$$

where M_i is the molar mass of component i.

Starting from the phenomenological coefficients L_{ij}, the elements of a $(n-1) \times (n-1)$ matrix $\mathbf{\Delta}$ can be defined as [?]

$$\Delta_{ij} = (1 - x_i) \left(\frac{L_{ij}}{x_j} - \frac{L_{in}}{x_n} \right) - x_i \sum_{k=1 \neq i}^{n} \left(\frac{L_{kj}}{x_j} - \frac{L_{kn}}{x_n} \right), \tag{6}$$

so that its inverse matrix $\boldsymbol{B} = \mathbf{\Delta}^{-1}$ is related to the Maxwell-Stefan diffusion coefficients $Đ_{ij}$. In the case of a quaternary mixture, the six Maxwell-Stefan diffusion coefficients are then given by [?]

$$Đ_{14} = \frac{1}{B_{11} + (x_2/x_1)B_{12} + (x_3/x_1)B_{13}}, \tag{7}$$

$$Đ_{24} = \frac{1}{B_{22} + (x_1/x_2)B_{21} + (x_3/x_2)B_{23}}, \tag{8}$$

$$Đ_{34} = \frac{1}{B_{33} + (x_1/x_3)B_{31} + (x_2/x_3)B_{32}}, \tag{9}$$

$$Đ_{12} = \frac{1}{1/Đ_{24} - B_{21}/x_2}, \tag{10}$$

$$Đ_{13} = \frac{1}{1/Đ_{14} - B_{13}/x_1}, \tag{11}$$

$$Đ_{23} = \frac{1}{1/Đ_{24} - B_{23}/x_2}. \tag{12}$$

$$\tag{13}$$

MD simulation runs for Lennard-Jones fluids were performed in order to test the validity of the quaternary Maxwell-Stefan diffusion coefficients calculated with *ms*2. For this purpose, a quaternary Lennard-Jones pseudo mixture was created with different name labels only for the molecules divided

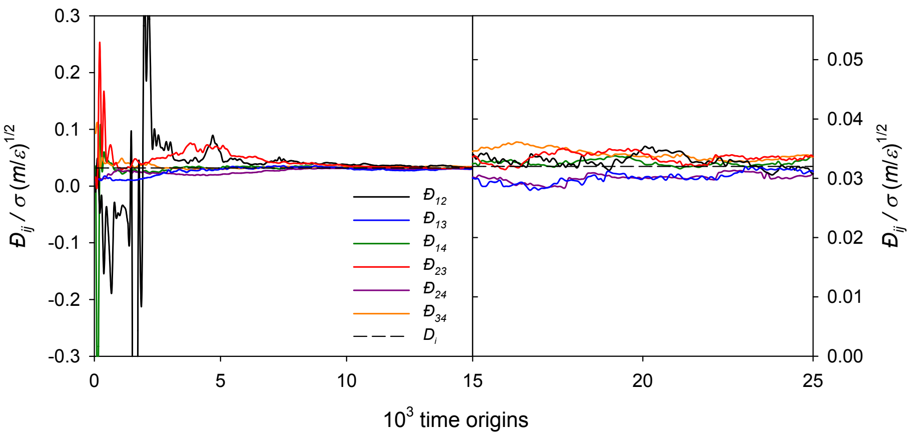

Figure 3: Maxwell-Stefan diffusion coefficients $Đ_{ij}$ as a function of time origins for a quaternary pseudo mixture ($\epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon_4$ and $\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$) at $k_\mathrm{B}T/\epsilon_1 = 0.728$ and $\rho\sigma_1^3 = 0.8442$.

into four mole fractions (x_1= 0.1, x_2= 0.2, x_3 = 0.3 and x_4 = 0.4 mol mol^{-1}). A system containing 800 molecules was simulated in the dense liquid state and the resulting Maxwell-Stefan diffusion coefficients were compared with the corresponding self-diffusion coefficient that has to be identical in the present case.

Fig. 3 shows the development of the calculated values of the six different Maxwell-Stefan diffusion coefficients with the number of time origins. As can be seen, the resulting values become independent after around 10^4 time origins and then oscillate around their mean value. The higher statistical uncertainties of $Đ_{12}$ and $Đ_{13}$ are due to the small number of species 1 molecules compared with the number of molecules of the other species.

6. Hydrogen bonding

There is no unambiguous characterization of a hydrogen bond between two molecules [**?** **?** **?**]. Rather, the hydrogen (H) bond 'is a structural motif and involves at least three atoms' [**?**]. The International Union of Pure and Applied Chemistry (IUPAC) defines the hydrogen bond as 'an attractive interaction between a hydrogen atom from a molecule or molecular fragment X–H in which X is more electronegative than H, and an atom or a group of atoms in the same or a different molecule, in which there is evidence of bond formation' [**?**].

Energetic [**?**], geometric [**?**] and topological [**?**] hydrogen bonding criteria have been proposed in the literature. Geometric criteria, which are not overly complex, are often based on the following assumptions [**?**]:

- The interaction between two hydrogen bonded sites is highly directional and short ranged.
- A donor interacts at most with a single acceptor, an acceptor may interact with multiple donors.

Accordingly, a class of geometric criteria for the evaluation of hydrogen bonding networks in fluids was implemented in *ms*2. Thereby, the triangle between three charge sites, being part of two different molecules, is evaluated to determine whether an interaction between two sites constitutes a hydrogen bond or not. A molecule acts as a donor to another molecule, i.e. the acceptor, if the following conditions hold [**?** **?** **?** **?**], cf. Fig. 4:

- The distance between the donor and the acceptor is smaller than a threshold distance, i.e. ℓ_{AD} or ℓ_{DA}.
- The distance between the acceptor sites of the acceptor and donor molecules is smaller than ℓ_{AA}.
- The angle between the acceptor-donor axis and the acceptor-acceptor axis is smaller than a threshold angle, i.e. φ_{DAA} or φ_{AAD}.

Therein, ℓ_{AD}, ℓ_{DA}, ℓ_{AA}, φ_{DAA} and φ_{AAD} are the parameters of the implemented class of hydrogen bonding criteria. For methanol, e.g., Haughney *et al.* [**?**] proposed $\ell_{\mathrm{AD}} = \ell_{\mathrm{DA}} = 2.6$ Å, $\ell_{\mathrm{AA}} = 3.5$ Å and $\varphi_{\mathrm{AAD}} = \varphi_{\mathrm{DAA}} = 30°$. Such geometric criteria have been applied to a variety of fluids, in particular to water [**?** **?** **?**], methanol-water mixtures [**?** **?**] or ethanol [**?** **?** **?**]. Hydrogen bonds in sorbate-sorbent interactions can be treated analogously [**?**]. E.g., for hydrogen fluoride, a simpler distance-based criterion can be used [**?**], which is also covered by the present implementation. A detailed description of the parameter setup is given in the supplementary material.

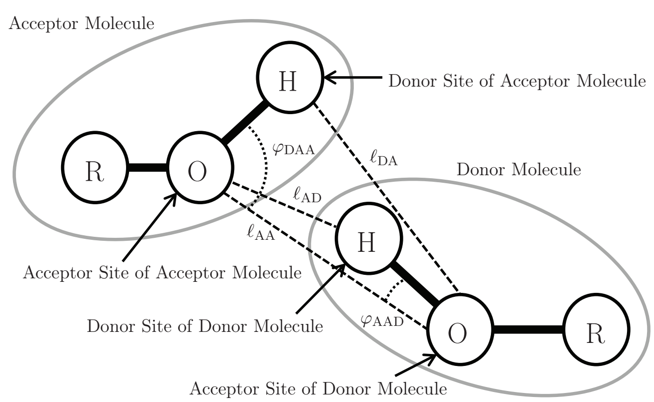

Figure 4: Hydrogen bonding criteria implemented in the present release version of *ms*2.

To to test the capabilities of the hydrogen bonding statistics in *ms*2 a MD simulation run of the ternary mixture water + methanol + ethanol (x_1 = 0.33 mol mol^{-1}, x_2 = 0.33 mol mol^{-1}, x_3 = 0.34 mol mol^{-1}) at T = 298.15 K and p = 0.1 MPa was performed with 4000 molecules. In this mixture all species can form hydrogen bonds with each other because all three molecules contain hydroxyl groups. Thus the hydrogen atoms may act as donors and the oxygen atoms as acceptors to form hydrogen bonds. Throughout, between like and unlike molecules, the geometric criteria of Haughney *et al.* [**?**] were used. The results are listed in Table 3. The hydrogen bonding statistics in *ms*2 not only indicates the amount of monomers (no bond), dimers (one bond), trimers (two

Table 3: Hydrogen bond statistics of the ternary mixture water (w) + methanol (m) + ethanol (e) (x_w = 0.33 mol mol^{-1}, x_m = 0.33 mol mol^{-1}, x_e = 0.34 mol mol^{-1}) at T = 298.15 K and p = 0.1 MPa in relative terms.

	water	methanol	ethanol
monomer	0.1%	0.9%	1.1%
dimer	1.2%	10.6%	11.5%
bonded to (w)	0.8%	3.5%	4.1%
(m)	0.2%	3.3%	3.6%
(e)	0.2%	3.8%	3.8%
trimer	7.8%	47.7%	42.6%
bonded to (w)(w)	2.8%	2.8%	2.8%
(w)(m)	1.9%	10.5%	8.9%
(w)(e)	2.0%	12.3%	10.2%
(m)(m)	0.3%	5.0%	4.9%
(m)(e)	0.4%	11.0%	10.3%
(e)(e)	0.4%	6.1%	5.5%
tetramer	25.8%	35.9%	36.8%
bonded to (w)(w)(w)	3.9%	15.6%	12.7%
(w)(w)(m)	4.9%	7.6%	7.8%
(w)(w)(e)	5.1%	7.7%	7.6%
(w)(m)(m)	2.7%	0.7%	1.3%
(w)(m)(e)	3.8%	1.7%	2.8%
(w)(e)(e)	2.8%	0.9%	1.5%
(m)(m)(m)	0.6%	0.2%	0.4%
(m)(m)(e)	0.7%	0.6%	1.1%
(m)(e)(e)	0.6%	0.7%	1.2%
(e)(e)(e)	0.7%	0.2%	0.4%
4 or more bonds	65.1%	4.9%	8.0%

bonds) and tetramers (three bonds), but also provides information about which molecule species are bonded.

7. OPAS Method

The OPAS (osmotic pressure for the activity of the solvent) method was implemented in *ms*2. It is an alternative to standard methods, e.g. the Widom test particle method [?] or thermodynamic integration [? ? ?], for calculating chemical potentials in the liquid phase in MD simulations. It is particularly well-suited for studying the solvent activity in electrolyte solutions, but can in principle also be used for studying mixtures of molecular species. Details and a thorough assessment of the method are presented in Ref. [?], the method is only briefly outlined in the following.

The basic idea is a direct simulation of the osmotic equilibrium between a pure solvent phase and a solution phase by introducing semi-permeable membranes in the simulation volume. These are realized by an external force field that acts only on the solute molecules to keep them in the solution phase. By sampling the total net membrane force per membrane area, the pressure difference between the two phases, called the osmotic pressure Π, is monitored. Assuming an incompressible solvent, the solvent activity is related to this osmotic pressure by

$$\ln a_\mathrm{s} = -\frac{\Pi v_\mathrm{s}}{RT}, \tag{14}$$

where v_s is the molar volume of the pure solvent at the temperature T, which is easily available from separate, standard molecular simulation runs. If an electrolyte solution is considered, the activity coefficient of the salt can be obtained by performing OPAS simulations at various salt molalities and applying the Gibbs-Duhem equation to the results for the solvent activity.

Fig. 5 shows molecular simulation results for the mean ionic activity coefficient of NaCl in aqueous solution at $T = 298.15$ K. Therein, results obtained by different groups employing different computational approaches are presented. All groups studied the molecular models by Joung and Cheatham [**?**] for Na^+ and Cl^- ions together with the SPC/E water model, and all simulation results are in mutual agreement.

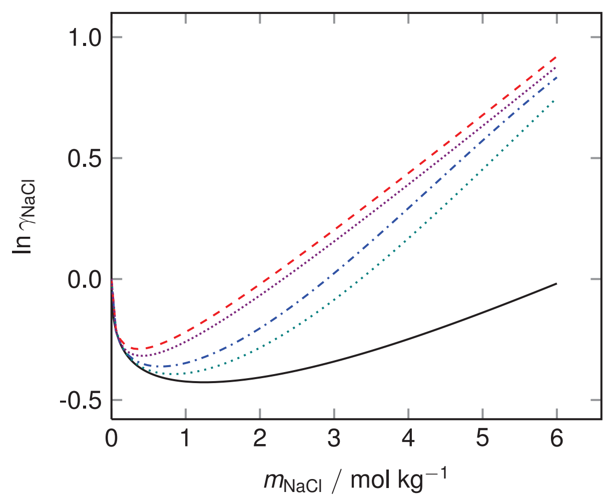

Figure 5: Mean ionic activity coefficient of NaCl over salt molality for aqueous NaCl solutions at $T = 298.15$ K using the SPC/E + Joung-Cheatham [**?**] model combination. The colored lines represent simulation results by different groups for that model combination (from top to bottom): OPAS simulations by Kohns *et al.* [**?**] (red dashed line), free energy calculations by Benavides *et al.* [**?**] (violet densely-dotted line), gradual insertion of ion pairs by Mester and Panagiotopoulos [**?**] (blue dashed-dotted line), and osmotic ensemble Monte Carlo simulations by Moučka *et al.* [**?**] (green dotted line). The black solid line shows the correlation to experimental data by Hamer and Wu [**?**].

8. Electrostatic long-range corrections

The applicability of *ms*2 was extended to electrically charged molecules or ions. The long ranged intermolecular interactions between those molecules and the surrounding solvents are considered using two well established approaches, Ewald summation and smooth-particle mesh

Ewald summation (SPME). Both approaches are well known [?] and are introduced only briefly here.

In Ewald summation, the overall coulombic potential acting in the simulation volume is determined by a sum of two terms, the short range coulombic contribution $u^{\mathrm{c,short}}$ and the long range coulombic contribution $u^{\mathrm{c,long}}$ according to

$$u^{\mathrm{c}} = u^{\mathrm{c,short}} + u^{\mathrm{c,long}} \ . \tag{15}$$

The separation into these terms is achieved by the introduction of a fictive charge density function $\rho^{\mathrm{screen}}(\boldsymbol{r})$, which acts in the entire simulation volume.

Following the Ewald summation approach, for any configuration of molecules in the simulation volume, each point charge of magnitude q_l in the simulation volume is superimposed with one countercharge of magnitude $-q_l$. Due to the presence of this superimposed charge, the interaction potential decays to zero within a distance reasonably low for molecular simulation and can, hence, be explicitly considered by the short term contribution of the Ewald summation.

The second term in Eq. (15) determines the contributions that were subtracted due to the introduction of the fictive charge density function. This term cannot be determined explicitly by the evaluation of pairwise interactions, since the coulombic potential at $r_{lm} = L/2$, i.e. the largest distance accessible in molecular simulation, still exhibits significant contributions to the potential energy of the system. In Ewald summation, these contributions are determined in Fourier space from the negative charge distribution function $-\rho^{\mathrm{screen}}(\boldsymbol{r})$. Since $\rho^{\mathrm{screen}}(\boldsymbol{r})$ depends on molecular motions and hence molecular positions, a Fourier transformation is performed for every configurational change in the simulation volume.

The SPME is widely considered as an improved Ewald summation method. In this approach, the concept of splitting the long range charge-charge interactions is fully employed. The difference between both methods lies only in the calculation of the long range contributions. In the SPME approach, the electrostatic field of ions in the simulation volume is described by a spline function with a given functional form. For this given spline equation, the Fourier transformation is known and has, hence, not to be determined in each simulation step. This accelerates the calculation of the long range contribution and reduces simulation time and effort. The accuracy of the SPME results is equivalent to the Ewald approach.

9. Vectorization

Essential for the vectorization of loops is how accessed data is distributed in memory. Deducing this information automatically from the code can be very difficult for the compiler, if there is indirection. In *ms*2, many arrays are accessed via array pointers and the order of the underlying data is therefore obfuscated. The most relevant information is wether or not data is contiguous in memory. To provide this information to the compiler directly, array pointers can be given the attribute "contiguous". The FORTRAN standard specifies:

> The contiguous attribute specifies that [...] an array pointer can only be pointer associated with a contiguous target.

This means that the array elements of a contiguous array pointer are not separated by any other data, potentially enabling a higher degree of vectorization. In version 3.0 of $ms2$, array pointers associated with contiguous targets have been given the contiguous attribute. The sequential performance gains – and the resulting parallel performance gains – achieved through this optimization depend on the specific simulation scenario, but are on average very significant. For MC and MD simulations, a sweet of simulations was run to evaluate the performance gains. The sweet covered the NPT and NVT ensembles, thermodynamic integration, Widom, different thermostats, different pure substances and mixtures. A total of 71 simulations were performed. For MC simulations, the observed average reduction of runtime was more than 7% and for MD simulations more than 20%.

10. Parallel ensemble calculations

$ms2$ was already parallelized in its initial version for distributed memory architectures using MPI [**? ?**]. The present version adds a new level of parallelization. The calculation of different ensembles is independent of each other. Therefore sampling different state points can be done concurrently. To achieve this, the processing elements (PE) were split in c disjunct groups such that each group consecutively computes an ensemble that is assigned to it. To enable this feature, the *mpiEnsembleGroups* option was introduced in the input file. The default (if not set or set to 0) is not to split the PE and to use a single MPI group. A value of 1 will enable the feature and automatically set the number of MPI groups to a minimum of the number of ensembles and the number of PE. Otherwise, this option will set the number of MPI groups according to the specified value. Another change applies to the restart capability of $ms2$. A checkpoint now consists of multiple restart (*.rst) files and can be restarted with the -r (or –restart) option of $ms2$.

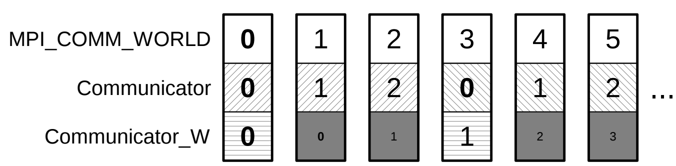

Figure 6: $ms2$ MPI ranks

$ms2$ creates a MPI communicator for each of the aforementioned MPI groups with MPI_Comm_Split (see Figure 6). The "coloring", i.e. the assignment of the PE to the MPI communicators, is done in continguous blocks. This is advantageous compared to e.g. a round robin fashion if the processes have to be pinned to NUMA domains. Another communicator contains the root processes (subcommunicator rank 0 processes) of all groups to ease the communication on a higher level among the groups (and there still is MPI_COMM_WORLD).

Even if the computation of different ensembles is embarrassingly parallel, there is still an interaction between the different communicators, e.g. when the program receives a signal to terminate and write a checkpoint before terminating. This signal is received from an arbitrary single PE (or non-isochronic from multiple PE). In the preceding releases of $ms2$, where ensembles were calculated sequentially with a single global communicator, a MPI_Allreduce call took care to spread this information among the processes after each iteration and is still acceptable within the subcommunicators. A collective communication of all PE beyond subcommunicator boundaries for each iteration in the computation would also implicitly synchronize all ensemble computations at this point. This is not a satisfactory solution, since computationally less intensive ensembles will be forced to wait for the slower ones. The present implementation uses non-blocking communication between the subcommunicator roots to avoid this.

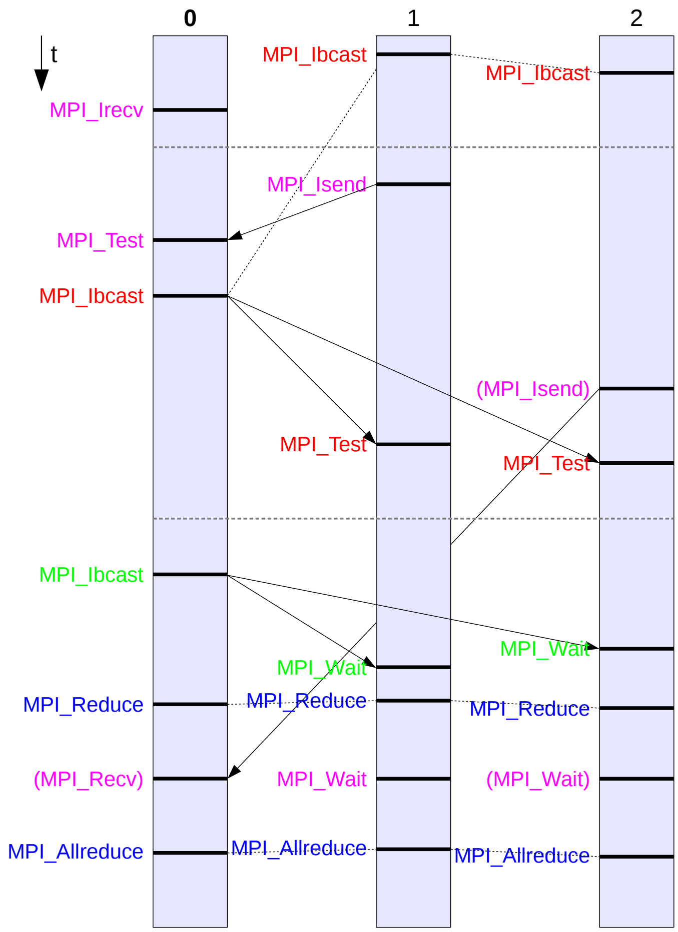

Figure 7: $ms2$ communication to pass termination status to all PEs

Before the iterations start, the root process will execute a MPI_Irecv, whereas all other processes start a MPI_Ibcast (see Figure 7). During the iterations, any process may trigger a termination by sending a message to the root, which will broadcast the information to notify all other processes. If no termination occurs during the iterations, the root will send a message to itself and broadcast a non-termination message to all other processes to finish the outstanding receive and broadcast. To take care of several processes sending a termination message at once, a summation reduction determines the sum of all respective messages for the root process to receive them all. This technique can also be used hierarchically, replacing the MPI_Allreduce call within each subcommunicator.

After a termination, every subcommunicator writes its own restart (*.rst) file.